

# Daniel Ng

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## Education

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| <b>Purdue University</b>                                                                                                                                                                                                                                                                    | West Lafayette, IN  |
| Bachelor of Science in Computer Science and Data Science, Graphics Concentration                                                                                                                                                                                                            | Aug 2024 - May 2028 |
| <ul style="list-style-type: none"><li>Cumulative GPA: 3.83/4.0 (4x Dean's List)</li><li>Relevant Coursework: Operating Systems, Fundamentals of Computer Graphics, Numerical Methods, Systems Programming, Data Structures and Algorithms, Linear Algebra, Multivariable Calculus</li></ul> |                     |

## Experience

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| <b>Purdue Envision Center - Rosen Center for Advanced Computing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | West Lafayette, IN  |
| Software Engineer (Realtime Simulation)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mar 2026 - Present  |
| <ul style="list-style-type: none"><li>Developed 6+ core features for an interactive Unity-based fluid dynamics laboratory exploring high-resolution scientific datasets</li><li>Eliminated 100% of runtime crashes caused by unbounded heap growth by designing a fixed-buffer memory allocation system in C#, resulting in stable memory usage during repeated loading of 100+ MB datasets</li><li>Collaborated with staff engineers to develop CollabXR, an open-source mixed-reality classroom, by improving application stability, developing workflows for publishing 3D environments, and documenting setup for users and developers</li><li>Restored reliable asset loading by reverse-engineering and refactoring legacy asynchronous pipelines, tracing data flow from AWS S3 through Unity runtime systems</li></ul> |                     |
| <b>Purdue Data Mine Corporate Partners Program (Caterpillar Inc.)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | West Lafayette, IN  |
| Undergraduate Research Software Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Aug 2025 - May 2026 |
| <ul style="list-style-type: none"><li>Led a 4-5 member subteam developing the REST API backend for an AI-powered machine data analysis platform using Python, FastAPI, AWS Bedrock, and Docker</li><li>Architected and implemented an MCP server exposing 5+ custom data analysis and visualization tools, including statistical summaries, histograms, time-series graphs, and geospatial mapping</li><li>Developed an LLM orchestration pipeline to automate multi-step analysis and visualization from natural-language queries</li></ul>                                                                                                                                                                                                                                                                                   |                     |
| <b>Fallingdani Games</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | New York, NY        |
| Independent Game Developer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Jul 2023 - Present  |
| <ul style="list-style-type: none"><li>Developed 11 Unity/C# games both independently and with teams, working across programming, design, art, and audio</li><li>Shipped multiple releases under strict 2-3 day competition deadlines by rapidly prioritizing features and managing scope</li><li>Placed top 3% in enjoyment out of 9.5k entries at GMTK Game Jam 2025 by developing novel mechanics for a twin-stick shooter</li><li>Reached 1,700+ views and 700+ plays across 11 total released projects on itch.io</li></ul>                                                                                                                                                                                                                                                                                                |                     |

## Projects

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| <b>Falling Engine - Data-Oriented Game Engine</b>   C++, OpenGL, GLSL, GLFW, CMake                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026 |
| <ul style="list-style-type: none"><li>Built a data-oriented C++ game engine with a sparse-set entity-component system (ECS), asset database, and OpenGL renderer, sustaining 60+ FPS with 36,000+ simulated entities and 32,000+ rendered meshes</li><li>Designed a cache-friendly ECS with paged sparse storage and dense component arrays, enabling constant-time component lookup</li><li>Developed a lazy asset pipeline for dependency-linked glTF models with persistent metadata and stable asset references across imports and reloads</li><li>Decoupled rendering from gameplay systems through a graphics-device abstraction with GPU resource caching for vertex buffers, textures, and shader programs</li></ul> |      |
| <b>NYC Service Watch - Data Engineering &amp; Analytics Pipeline</b>   Python, PostgreSQL, dbt, Streamlit, Docker, GitHub Actions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026 |
| <ul style="list-style-type: none"><li>Built an end-to-end ELT pipeline for NYC service request data with incremental ingestion through Polars, PostgreSQL, and dbt</li><li>Improved warehouse reliability by implementing automated dbt quality tests, layered models, and pre-aggregated analytics marts</li><li>Developed a Streamlit dashboard with sample data and visualizations to validate ingestion and transformed warehouse outputs</li><li>Automated CI validation with GitHub Actions and Docker by provisioning databases, running tests, and verifying dashboard startup</li></ul>                                                                                                                             |      |
| <b>Echoes of Isovios - 2D Roguelike Platformer</b>   Unity Engine, C#, Git                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025 |
| <ul style="list-style-type: none"><li>Developed 85% of the enemy roster, including 17 enemy classes and all 4 boss battles, for a Steam-published game using Unity, C#, finite state machines and animation events</li><li>Refactored enemy architecture to separate shared logic from specialized behaviors, enabling faster prototyping and balancing</li><li>Collaborated with a 20+ member multidisciplinary team using Git/GitHub to design, develop and ship a complete game</li></ul>                                                                                                                                                                                                                                 |      |

## Technical Skills

**Languages:** C++, C, C#, GLSL, Python, SQL, Java, R, HTML/CSS  
**Frameworks/Libraries:** OpenGL, GLM, GLFW, FastAPI, Streamlit, PyTorch, NumPy  
**Tools/Platforms:** Unity, Docker, PostgreSQL, Git/GitHub, GitHub Actions, CMake, AWS, Ollama